

d) $F > O > C > N$

11. The oxide which cannot act as a reducing agent is
a) NO_2 b) CO_2 c) SO_2 d) ClO_2

KATHMANDU MODEL SECONDARY SCHOOL
Second Term Examination- 2080
SET-A

Class: XI 'Science'
Time: 25 mins
Subject: Com. Mathematics (0071)
Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

GROUP 'A' MCQ [11x1=11]

Re-write the correct option of each question in your answer sheet.

1. If A and B are two sets then $A \cap (A \cup B)^c$ is equal to

- a) $\emptyset$ b) B c) $\emptyset$ d) A - B

2. $|x+2| \leq 1$ is equivalent to

- a) $1 \leq x \leq \frac{1}{3}$ b) $1 \leq x \leq \frac{1}{3}$ c) $-1 \leq x \leq -\frac{1}{3}$ d) $-\frac{1}{2} \leq x \leq \frac{1}{3}$

3. A function $f: N \rightarrow N$ is defined by $f(x) = 2x + 1$, for all $x \in N$, then f is

- a) One to one but not onto function b) onto but not one to one function
c) One to one and onto both function d) neither one to one nor onto function

4. Sum of the infinite Geometric series: $(-\frac{5}{4} + \frac{5}{16} - \frac{5}{64} + \dots)$ is

- a) $-\frac{1}{4}$ b) $-\frac{1}{4}$ c) $\frac{5}{4}$ d) -1

5. Distance between the lines $2x - 5y - 6 = 0$ and $6x - 15y + 11 = 0$ is

- a) 17 b) 17 c) $\frac{29}{\sqrt{261}}$ d) $\frac{17}{\sqrt{261}}$

6. $\frac{2-x}{x-2}$ is

- a) 1 b) 1 c) -1 d) doesn't exist

7. If t and $y = \frac{1}{t}$ then $\frac{dy}{dx}$ is

- a) $\frac{y}{x}$ b) $\frac{y}{x}$ c) $-\frac{y}{x}$ d) $-\frac{x}{y}$

8. The value of 'a' the function $f(x) = \begin{cases} ax - 5 & \text{if } x > 2 \\ x - 1 & \text{if } x \leq 2 \end{cases}$ is continuous at $x = 2$?

- a) 1 b) -2 c) 1 d) 3

9. The value of $\lim_{x \rightarrow 0} \frac{e^{2x} - 1}{\ln(1+x)}$ is

- a) 1 b) 1 c) 2 d) doesn't exist

10. The value of $(1+i)^4 (1+\frac{1}{i})^4$ is

- a) 8 b) 8 c) 16 d) 32

11. The lines given by $ax^2 + 2hxy + by^2 = 0$ are real and distinct if

- a) $h^2 < ab$ b) $h^2 < ab$ c) $h^2 = ab$ d) $a + b = 0$

Note: The question set must be written in the answer book

KATHMANDU MODEL SECONDARY SCHOOL
Second Term Examination- 2080
SET-B

Class: XI 'Science'
Time: 25 mins
Subject: Com. Mathematics (0071)
Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

GROUP 'A' MCQ [11x1=11]

1. If A = (-1, 3) and B = [0, 4] then A - B is

- a) (-1, 0) b) [-1, 0) c) (-1, 0) d) [-1, 0]

2. A function $f: [-2, 2] \rightarrow R$, defined by $f(x) = x^2$ is

- a) One to one but not onto function b) onto but not one to one function
c) One to one and onto both function d) neither one to one nor onto function

3. The simplest form of $(\frac{1-i}{1+i})^3$ is

- a) 0 b) 1 c) 3i d) i

4. The value of $\lim_{x \rightarrow 0} \frac{\sin x}{x}$ is

- a) 0 b) 1 c) -1 d) doesn't exist

5. The derivative of $f(x) = (3-x)^5$ is

- a) $5(3-x)^4$ b) $(3-x)^4$ c) $-5(3-x)^4$ d) $5(3-x)^5$

6. If A and B are two sets then $A \cap (B - A)$ is equal to

- a) A b) B c) $\emptyset$ d) none

7. The sum of the infinite Geometric series: $(-\frac{5}{4} + \frac{5}{16} - \frac{5}{64} + \dots)$ is

- a) $\frac{1}{4}$ b) $-\frac{1}{4}$ c) $\frac{5}{4}$ d) -1

8. The lines given by $ax^2 + 2hxy + by^2 = 0$ are real and distinct if

- a) $h^2 > ab$ b) $h^2 < ab$ c) $h^2 = ab$ d) $a + b = 0$

9. The distance between the lines $2x - 5y - 6 = 0$ and $6x - 15y + 11 = 0$ is

- a) 29 b) 17 c) $\frac{29}{\sqrt{261}}$ d) $\frac{17}{\sqrt{261}}$

10. For what value of 'a' the function $f(x) = \begin{cases} ax - 5 & \text{if } x > 2 \\ x - 1 & \text{if } x \leq 2 \end{cases}$ is continuous at $x = 2$?

- a) 4 b) -2 c) 1 d) 3

11. The value of $\lim_{x \rightarrow 0} \frac{e^{2x} - 1}{\ln(1+x)}$ is

- a) 0 b) 1 c) 2 d) doesn't exist

Note: The question set must be written in the answer book

KATHMANDU MODEL SECONDARY SCHOOL
Second Term Examination- 2080

XI 'Science'

Subject: Com. Mathematics (0071)

F.M.: 75

P.M.: 30

Answers are required to give their answers in their own words as far as practicable.
Answers in the margin indicate full marks.

GROUP 'B' [8×5=40]

Define power set with example. [1+1+3]

For any two sets A and B

Prove that $A - B = \overline{B} - \overline{A}$

For any sets A, B and C

Prove that $A - (B \cap C) = (A - B) \cup (A \cap C)$

Define absolute value of real number. Interpret it geometrically. If 'a' be

positive real number and $x \in \mathbb{R}$,

Prove that: $|x| \leq a \iff -a \leq x \leq a$. [1+1+3]

Define one to one function.

If $f: \mathbb{R} - \{3\} \rightarrow \mathbb{R} - \{1\}$ given by $f(x) = \frac{x}{x-3}$.

Show that f is bijective function. Also find $f^{-1}(x)$ [1+3 +1]

Define continuity of a function at a given point.

Function defined by

$$f(x) = \begin{cases} x^2 + 2 & \text{for } x < 5 \\ 20 & \text{for } x = 5 \\ 3x + 12 & \text{for } x > 5 \end{cases}$$

Does the limit of $f(x)$ exist at $x = 5$?

Is the function $f(x)$ continuous at $x = 5$?

If not, state the type of discontinuity? [1+1+1+1+1]

How can you make it continuous?

Two equations of straight lines are given by

$$L_1: a_1x + b_1y + c_1 = 0$$

$$L_2: a_2x + b_2y + c_2 = 0$$

Write the condition that the above lines L_1 and L_2 are perpendicular. [1]

Write the expression for the perpendicular distance from point [1]

(x_1, y_1) to line L_1 [3]

If z and w are two complex number, then prove that:

$$|z + w|^2 = |z|^2 + |w|^2 + 2\operatorname{Re}(z \cdot \bar{w}) \quad [2]$$

$$\text{Evaluate: } \lim_{x \rightarrow 0} \frac{\sqrt[3]{1+x} - \sqrt[3]{1-x}}{x} \quad [3]$$

$$\text{Evaluate: } \lim_{y \rightarrow 0} \frac{(x+y)\sec(x+y) - x\sec x}{y}$$

10. In the electrolysis of fused salt, the amount of the substance deposited on electrode will not depend on
a) current intensity
c) temperature of the bath
11. The oxide which cannot act as a reducing agent is
a) NO_2
b) CO_2
c) SO_2
d) ClO_2

18. a) Find the value of 'k' for which the roots of the equation. [2]

$$x^2 - kx + (k+1) = 0 \text{ are in the ratio } 2:3.$$

b) Find the condition that the two quadratic equation

$$ax^2 + bx + c = 0 \text{ (} a \neq 0 \text{) and } a_1x^2 + b_1x + c_1 = 0 \text{ (} a_1 \neq 0 \text{) have one root common. [3]$$

19. a) The AM between two numbers exceeds their GM by 2 and GM exceeds the HM by 1.6. Find the two numbers. [2]

b) A rubber Ball is dropped from a Height of 10m. If it rebounds approximately one half the distance after each fall. Use infinite geometric series to approximate the total distance the ball travels before coming to rest. [3]

GROUP 'C' [3×8=24]

20. a) Solve the inequality:

$$\frac{x^2 - 2x}{x-1} \leq 0 \quad [2]$$

b) Construct the truth table for compound statement [2]

$$\sim(p \wedge q) \iff (\sim p \vee \sim q).$$

c) Find square root of complex number [2]

$$\frac{5+12i}{3-4i}$$

d) Find the quadratic equation with rational coefficients whose one of the

$$\text{root is } \frac{1}{5+3i}$$

[1+2+2]

21. a) Find $\frac{dy}{dx}$ if

$$\text{i) } y = \frac{f(x)}{g(x)}, \quad (g(x) \neq 0) \quad \text{ii) } y = \sqrt{\frac{x}{a+x}} \quad \text{iii) } x^3 + y^3 - 3xy^2 = 0$$

b) Find differential coefficient of $\sqrt{\tan 2x}$ by using first principle. [3]

22. A general equation of second degree is given by

$$18x^2 + 6xy - 4y^2 - 15x - 40y - 75 = 0 \dots\dots\dots \text{eq}^n(1)$$

i) Prove that $\text{eq}^n(1)$ represents a pair of straight lines. [1]

ii) Find separate equation of lines given by $\text{eq}^n(1)$. [2]

iii) Find the angle between the lines given by $\text{eq}^n(1)$. [1]

iv) Find equation of bisectors of angle between the lines obtained by $\text{eq}^n(1)$. [1]

v) Identify the acute angle bisectors. [1]

vi) Identify the bisectors that contain the origin. [1]

Best of Luck!

- iv) Find the quadratic equation whose one root is $-2 + \sqrt{3}$
 v) Find the value of $3^{1/3} \cdot 3^{1/9} \cdot 3^{1/27} \dots$

[1]

19. a) Evaluate: $\lim_{x \rightarrow \frac{\pi}{2}} \frac{e^{\cos x} - 1}{\frac{\pi}{2} - x}$

[1]

[2]

- b) If the quadratic equations $x^2 + px + q = 0$ and $x^2 + p'x + q' = 0$ have a common root then prove that the root must be either $\frac{pq' - p'q}{q - q'}$ or $\frac{q - q'}{p' - p}$

[3]

GROUP-B [3×8=24]

20. a) A square is drawn by joining the mid points of the sides of a given square. A third square is drawn inside the second square in the same way and this process continues indefinitely. If a side of the first square is 16 cm, determine the sum of the areas of all the squares.

[3]

- b) The sum of an infinite number of terms of a GP is 23 and sum of their square is 69. Determine the first term and the common ratio.

[2]

- c) If the roots of the equation $ax^2 + cx + c = 0$ are in the ratio $m:n$, then prove that: $\sqrt{\frac{m}{n}} + \sqrt{\frac{n}{m}} + \sqrt{\frac{c}{a}} = 0$

[3]

21. a) Define symmetric difference of two sets A & B. For any sets A and B, prove that $A \Delta B = (A \cup B) - (A \cap B)$

[1+3]

- b) Prove that for any two statements p and q

[2]

i) $\sim(p \vee q) \equiv \sim p \wedge \sim q$

[2]

ii) $\sim(p \vee (\sim q)) \equiv (\sim p) \wedge q$

22. Evaluate the following limits:

[2]

a) $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x}$

[2]

b) $\lim_{x \rightarrow \infty} \sqrt{x}(\sqrt{x+5} - \sqrt{x})$

[2]

c) $\lim_{\theta \rightarrow \frac{\pi}{4}} \frac{\cos \theta - \sin \theta}{\theta - \frac{\pi}{4}}$

[2]

- d) If $f(x) = \frac{ax+b}{x-1}$, $\lim_{x \rightarrow 0} f(x) = 2$ and $\lim_{x \rightarrow \infty} f(x) = 5$, find the value of $f(4)$

[2]

Best of luck!

- iv) Find the quadratic equation whose one root is $-2 + \sqrt{3}$ [1]
 v) Find the value of $3^{1/3} \cdot 3^{1/9} \cdot 3^{1/27} \dots$ [1]
 19. a) Evaluate: $\lim_{x \rightarrow \frac{\pi}{2}} \frac{e^{\cos x} - 1}{\frac{\pi}{2} - x}$ [2]
 b) If the quadratic equations $x^2 + px + q = 0$ and $x^2 + p'x + q' = 0$ have a common root then prove that the root must be either $\frac{pq' - p'q}{q - q'}$ or $\frac{q - q'}{p' - p}$ [3]

GROUP-B [3×8=24]

20. a) A square is drawn by joining the mid points of the sides of a given square. A third square is drawn inside the second square in the same way and this process continues indefinitely. If a side of the first square is 16 cm, determine the sum of the areas of all the square. [3]
 b) The sum of an infinite number of terms of a GP is 23 and sum of their square is 69. Determine the first term and the common ratio. [2]
 c) If the roots of the equation $ax^2 + cx + c = 0$ are in the ratio $m:n$, then prove that: $\sqrt{\frac{m}{n}} + \sqrt{\frac{n}{m}} + \sqrt{\frac{c}{a}} = 0$ [3]
21. a) Define symmetric difference of two sets A & B. For any sets A and B, prove that $A \Delta B = (A \cup B) - (A \cap B)$ [1+3]
 b) Prove that for any two statements p and q [2]
 i) $\sim(p \vee q) \equiv \sim p \wedge \sim q$ [2]
 ii) $\sim(p \vee (\sim q)) \equiv (\sim p) \wedge q$
22. Evaluate the following limits: [2]
 a) $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x}$ [2]
 b) $\lim_{x \rightarrow \infty} \sqrt{x}(\sqrt{x+5} - \sqrt{x})$ [2]
 c) $\lim_{\theta \rightarrow \frac{\pi}{4}} \frac{\cos \theta - \sin \theta}{\theta - \frac{\pi}{4}}$
 d) If $f(x) = \frac{ax+b}{x-1}$, $\lim_{x \rightarrow 0} f(x) = 2$ and $\lim_{x \rightarrow \infty} f(x) = 5$, find the value of $f(4)$ [2]

Best of luck!

Academic Excellence Through Quality Education
KATHMANDU MODEL SECONDARY SCHOOL
First Term Examination- 2080

Class: XI 'Science'
 Time: 3 hrs.

SET-A

Subject: Mathematics [0071]

F.M: 75

P.M: 30

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 The figures in the margin indicate full marks.

Attempt all the questions.

GROUP-A [MCQ] [11×1=11]

- For what value of k , the equation $(3k + 1)x^2 + 2(k + 1)x + k = 0$ may have reciprocal roots?
 a) $-\frac{3}{2}$ b) $-\frac{1}{2}$ c) $\frac{3}{2}$ d) 2
- If $A = \{a, b, c, d\}$ then total number of non-empty proper subsets of A is ...
 a) 4 b) 14 c) 15 d) 16
- Both roots of the equation $x^2 + ax + b = 0$ and $x^2 + dx + e = 0$ are common then
 a) $ab = de$ b) $d^2e = a^2b$ c) $ae = bd$ d) $ad = be$
- A and B are two non-empty sets then which one is correct?
 a) $(A \cup B) \subset (A \cap B)$ b) $(A \cap B) \subset (A \cup B)$
 c) $(A \cup B) = (A \cap B)$ d) $(A \cup B) - (A \cap B) = \emptyset$
- Let ' p ' be "10 is an even number" and ' q ' be "10 is multiple of 5" then symbolic representation of "10 is neither even nor multiple of 5" is
 a) $p \wedge q$ b) $(\sim p) \vee (\sim q)$ c) $p \vee q$ d) $(\sim p) \wedge (\sim q)$
- The value of $\lim_{x \rightarrow \infty} x \cdot \sin \frac{1}{x}$
 a) 1 b) -1 c) 0 d) doesn't exist
- $\lim_{x \rightarrow 0} \frac{\ln(x+1)}{e^x - 1} = ?$
 a) -1 b) 0 c) 1 d) doesn't exist
- The AM and HM between two number are 27 and 12 respectively then GM between two numbers be
 a) 18 b) 9 c) 36 d) 45
- The value of limit $\lim_{x \rightarrow 4} \frac{x^2 - 2x - 8}{x - 4}$ is
 a) 0 b) 2 c) 8 d) 6

- The compound statement $p \wedge (\sim p)$ is
 a) tautology b) contradiction c) implication d) inverse
- The sum of the infinite geometric series $1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} \dots$ is
 a) 0.5 b) 1 c) 1.5 d) 3

GROUP-B [8×5=40]

- a) Define subset of a set with example.
 b) For non-empty sets A, B and C and universal set U , prove that
 i) $\overline{A \cap B} = \overline{A} \cup \overline{B}$
 ii) $A - (B \cap C) = (A - B) \cup (A - C)$ [1+2+2]
- a) Define tautology and contradiction of statement with example.
 b) For any statement p and q , test whether $(p \wedge q) \wedge \sim(p \vee q)$ is a contradiction or not. [2+3]
- a) If one root of the equation $ax^2 + bx + c = 0$ ($a \neq 0$) be square of the other, then prove that $b^3 + a^2c + ac^2 = 3abc$ [2]
 b) Under what conditions the roots of quadratic equation $ax^2 + bx + c = 0$ ($a \neq 0$) are
 i) real and unequal ii) imaginary iii) equal
 iv) equal in magnitude but opposite in sign
 c) For what value of k , the equation $9x^2 + kx + 1 = 0$ has equal roots? [1]
- a) The sum of three numbers in AP is 15. If 1, 4, 19 are added to them respectively, the resulting series is in GP. Find the numbers. [2]
 b) If a, b, c are in HP, then prove that $\frac{b+c}{bc}, \frac{a+c}{ac}, \frac{a+b}{ab}$ are in AP. [3]
- a) Define the limit of a function at a point.
 b) A function given by; $f(x) = \frac{x^2 - 16}{x - 4}$. Does $f(4)$ exist? Also find $\lim_{x \rightarrow 4} f(x)$ (if exist) [2]
 c) Evaluate: $\lim_{x \rightarrow 64} \frac{\sqrt[3]{x} - 2}{\sqrt{x} - 4}$ [2]
- Evaluate:
 a) $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}$ b) $\lim_{x \rightarrow y} \frac{x \tan x - y \tan y}{x - y}$ [2+3]
- i) What is the value of $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a}$ for all rational n and $n \neq 0$ [1]
 ii) If $A \subseteq B$ then show that $\overline{B} \subseteq \overline{A}$
 iii) Write the negation of the statement "If 10 is odd or 12 is even, then $5 > 7$ " [1]

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First Term Examination- 2080

Class: XI 'Science'
 Time: 3 hrs.

SET-B

Subject: Mathematics [0071]

F.M: 75
 P.M: 30

Candidates are required to give their answers in their own words as far as practicable.
 The figures in the margin indicate full marks.

Attempt all the questions.

GROUP-A [MCQ] [11×1=11]

- The AM and HM between two number are 27 and 12 respectively then GM between two numbers be
 a) 18 b) 9 c) 36 d) 45
- The sum of infinite geometric series $\frac{1}{2} + \frac{3}{8} + \frac{9}{32} + \dots$ is
 a) 1 b) 2 c) 4 d) doesn't exist
- The value of $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos x}{(\frac{\pi}{2} - x)}$
 a) 1 b) -1 c) 0 d) doesn't exist
- For what value of k , the equation $(3k+1)x^2 + 2(k+1)x + k = 0$ may have reciprocal roots?
 a) $-\frac{3}{2}$ b) $-\frac{1}{2}$ c) $\frac{3}{2}$ d) 2
- The value of limit $\lim_{x \rightarrow 4} \frac{\sqrt{x}-2}{x-4}$ is
 a) 0 b) $\frac{1}{4}$ c) $\frac{3}{4}$ d) None
- The compound statement $p \wedge (\sim p)$ is
 a) tautology b) contradiction c) implication d) inverse
- Both roots of the equation $x^2 + ax + b = 0$ and $x^2 + dx + e = 0$ are common then
 a) $ab = de$ b) $d^2e = a^2b$ c) $ae = bd$ d) $ad = be$
- $\lim_{x \rightarrow 0} x \cdot \sin \frac{1}{x}$ is
 a) 0 b) 1 c) ∞ d) doesn't exist
- Let 'p' be "10 is an even number" and 'q' be "10 is multiple of 5" then symbolic representation of "10 is neither even nor multiple of 5" is
 a) $p \wedge q$ b) $(\sim p) \vee (\sim q)$ c) $p \vee q$ d) $(\sim p) \wedge (\sim q)$

- If $A = \{a, b, c, d\}$ then total number of non-empty proper subsets of A is ...
 a) 4 b) 14 c) 15 d) 16
- If A and B are disjoint sets then which of the following is true?
 a) $A \subseteq \bar{B}$ b) $B \subseteq \bar{A}$
 c) $\bar{A} \subseteq \bar{B}$ d) $\bar{B} \subseteq \bar{A}$

GROUP-B [8×5=40]

- Define subset of a set with example.
 b) For non-empty sets A, B and C and universal set U, prove that
 i) $\overline{A \cap B} = \bar{A} \cup \bar{B}$
 ii) $\overline{A - (B \cap C)} = (A - B) \cup (A - C)$ [1+2+2]
- Define tautology and contradiction of statement with example.
 b) For any statement p and q, test whether $(p \wedge q) \wedge \sim(p \vee q)$ is a contradiction or not. [2+3]
- a) If one root of the equation $ax^2 + bx + c = 0$ ($a \neq 0$) be square of the other, then prove that $b^3 + a^2c + ac^2 = 3abc$ [2]
 b) Under what conditions the roots of quadratic equation $ax^2 + bx + c = 0$ ($a \neq 0$) are
 i) real and unequal ii) imaginary iii) equal [2]
 iv) equal in magnitude but opposite in sign
- For what value of k, the equation $9x^2 + kx + 1 = 0$ has equal roots? [1]
- a) The sum of three numbers in AP is 15. If 1, 4, 19 are added to them respectively, the resulting series is in GP. Find the numbers. [2]
 b) If a, b, c are in HP, then prove that $\frac{b+c}{bc}, \frac{a+c}{ac}, \frac{a+b}{ab}$ are in AP. [3]
- a) Define the limit of a function at a point. [1]
 b) A function given by; $f(x) = \frac{x^2-16}{x-4}$. Does f(4) exist? Also find $\lim_{x \rightarrow 4} f(x)$ (if exist) [2]
 c) Evaluate: $\lim_{x \rightarrow 64} \frac{\sqrt[6]{x}-2}{\sqrt[3]{x}-4}$ [2]
- Evaluate:
 a) $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}$ b) $\lim_{x \rightarrow y} \frac{x \tan x - y \tan y}{x - y}$ [2+3]
- i) What is the value of $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a}$ for all rational n and $n \neq 0$ [1]
 ii) If $A \subseteq B$ then show that $\bar{B} \subseteq \bar{A}$ [1]
 iii) Write the negation of the statement "If 10 is odd or 12 is even, then $5 > 7$ " [1]

Class: XI 'Science' Subject: Com. Mathematics (0071) F.M.: 75
Time: 3 hrs. P.M.: 30

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Attempt all the questions.

GROUP 'B' [8x5=40]

Short Answer Questions.

12. a) Write the contrapositive of the compound statement $q \Rightarrow p$. [1]
b) Write the negation of the statement "If 11 is odd or 2 is even then $5 < 7$ ". [2]

- c) Construct the truth table for the compound statement $(p \wedge q) \Rightarrow (p \vee q)$. [2]
Also, what conclusion can be drawn from the truth table? [2]

13. a) Define power set with an example. [2]
b) If A, B and C are subsets of a universal set U, prove that

i) $A \subseteq B \Rightarrow \bar{B} \subseteq \bar{A}$ [1]
ii) $A \subseteq B \Rightarrow C - B \subseteq C - A$ [2]

14. a) Find the values of the parameter a for which the quadratic equation $(a+1)x^2 + 2(a+1)x + (a-2) = 0$

- i) has two distinct roots [1]
ii) has no real roots [1]
iii) has two equal roots [1]

- b) If one root of the equation $x^2 - \sqrt{5}x - 19 = 0$ is $\frac{9+\sqrt{5}}{2}$, then find the other root [2]

15. a) If $a^x = b^y = c^z$ and a, b, c are in G.P then prove that x, y and z are in H.P. [3]
b) Under what condition sum of infinite geometric series exists? Give an example and find its sum. [2]

16. a) Under what condition $\sqrt{ab} = \sqrt{a} \times \sqrt{b}$ for a, b real numbers? Give an example to support this expression. [1]
b) Find the real and imaginary part of the complex number $\frac{2+\sqrt{-4}}{3+2i}$ [2]

- c) Find the values of following.

i) $i^{10} + i^{25} + i^{30} + i^{35}$ [1+1=2]
ii) $(-\sqrt{-1})^{4n+3}, n \in \mathbb{N}$

17. Evaluate:

a) $\lim_{x \rightarrow 4} \frac{x^2-16}{\sqrt{3x+4}-4}$ [2+3=5]
b) $\lim_{x \rightarrow 2} \frac{\sqrt{x-2}+\sqrt{x}-\sqrt{2}}{\sqrt{x^2-4}}$

18. Evaluate:
a) $\lim_{x \rightarrow 4} \frac{0^x - e^4}{\sqrt{x}-2}$
b) $\lim_{h \rightarrow 0} \frac{\sqrt{\sin(x+h)} - \sqrt{\sin x}}{h}$ [2+3=5]

19. a) Give an example of function which takes the form $\frac{0}{0}$, but whose limit does not exist. [2]
b) Is it possible to write $\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} = \lim_{x \rightarrow 0} x^2 \cdot \lim_{x \rightarrow 0} \sin \frac{1}{x}$? If not, give reason. [2]

- c) Evaluate:
i) $\lim_{x \rightarrow \infty} \frac{3x}{\sqrt{x^2+5}}$

GROUP 'C' [3x8=24]

20. a) Under what condition two compound statements are logically equivalent? Justify it by an example. [2]
b) Define Set. How does it differ from collection? Justify it by an example. [2]

- c) If A, B and C are subsets of universal set U, prove that:
i) $\overline{(A \cup B)} = \bar{A} \cap \bar{B}$
ii) $A - (B \cup C) = (A - B) \cap (A - C)$

21. a) Define identity. How does it differ from an equation? [2]
b) Under what condition roots of quadratic equation $px^2 + qx + r = 0$ are imaginary? Justify it by an example. [2]

- c) Derive the condition for two quadratic equations $p_1x^2 + q_1x + r_1 = 0$ and $p_2x^2 + q_2x + r_2 = 0$ to have one root common. [2]
d) If each term of an infinite geometric series is equal to three times the sum of all the terms which follow it and the sum of first two terms is 12, Find the sum to infinity of the series. [2]

22. a) Evaluate:

i) $(1+i)^6 \left(1 + \frac{1}{i}\right)^6$
ii) $\lim_{x \rightarrow y} \frac{x \sin x - y \sin y}{x - y}$

- b) If a, b, c are in A.P, b, c, a are in H.P then prove that c, a, b are in G.P. [2]
c) For a given function $f(x) = \frac{x-4}{\sqrt{x}-2}$.
i) Is it possible to find $f(2)$? Explain. [2]
ii) Does $\lim_{x \rightarrow 4} f(x)$ exist? Explain. [2]

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Attempt all the questions.

GROUP 'A' [MCQ] [11x1=11]

1. Density of NH_3 at 30°C and 5 atm pressure is 30 gm/lit. b) 2.43 gm/lit c) 3.42 gm/lit d) 3.42 gm/lit

2. Atmospheric air contains 20% oxygen by volume, p 2 mmHg b) 760 mmHg c) 380 mmHg d) 760 mmHg

3. Increase in volume that will take place when 1 mole of gas is heated from 300K to 360K at constant pressure is a) 20% b) 20% c) 20% d) 20%

4. In the following equation, the value of 'x' is $\text{S}_2\text{O}_8^{2-} + \text{X} \rightarrow \text{S}_2\text{O}_8^{2-} + \text{X}$ a) $\text{S}_2\text{O}_8^{2-}$ b) $\text{S}_2\text{O}_8^{2-}$ c) $\text{S}_2\text{O}_8^{2-}$ d) $\text{S}_2\text{O}_8^{2-}$

5. Electrolysis of fused salt, the amount of electrolyte will not depend on a) time b) temperature c) concentration d) nature of electrolyte

6. Which of the following cannot act as a reducing agent? a) CO_2 b) CO_2 c) SO_2 d) SO_2

7. Number of moles of Faraday needed to deposit 1 mole of copper is a) 2 b) 2 c) 3 d) 3

8. Most stable allotrope of sulphur is a) monoclinic b) plastic c) rhombic d) rhombic

9. Correct order of size is a) $\text{I}^- < \text{I}^+ < \text{I}$ b) $\text{I}^- < \text{I}^+ < \text{I}$ c) $\text{I}^- < \text{I}^+ < \text{I}$ d) $\text{I}^- < \text{I}^+ < \text{I}$

10. Correct order of ionization energy is a) $\text{N} > \text{O} > \text{C}$ b) $\text{N} > \text{O} > \text{C}$ c) $\text{N} > \text{O} > \text{C}$ d) $\text{N} > \text{O} > \text{C}$

Class: XI 'Science'

SET-B

Time: 25 min.

Subject: Com. Mathematics (0071)

Tick the best options among the four options. Apply tick on two options will be automatically desited and mark will not be counted.

GROUP 'A' MCQ [11×1=11]

Re-write the correct option of each question in your answer sheet.

1. What is the sum of the series $9 + 3 + 1 + \frac{1}{3} + \dots$?

A) $\frac{13}{2}$ B) 13 C) $\frac{25}{2}$ D) $\frac{27}{2}$

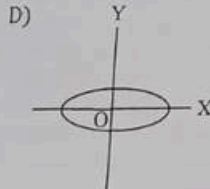
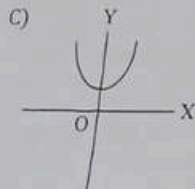
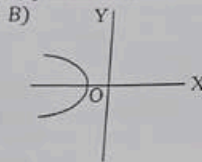
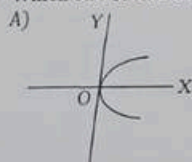
2. Which of the following compound statement is not true?

A) $\sim(p \wedge q) \Leftrightarrow [(\sim p) \vee (\sim q)]$ B) $\sim(p \vee q) \Leftrightarrow [(\sim p) \wedge (\sim q)]$
C) $\sim(p \Rightarrow q) \Leftrightarrow [p \wedge \sim q]$ D) $p \wedge (q \vee r) \Leftrightarrow [(p \vee q) \wedge (p \wedge r)]$

3. What is the solution of $|2 - x| \geq 5$?

A) $(-\infty, \infty)$ B) $(3, 7)$
C) $(-\infty, -3) \cup (7, \infty)$ D) $(-\infty, -3] \cup [7, \infty)$

3. Which one of the following graphs represents a function?



5. What is the multiplicative inverse of $z = (3, -2)$?

A) $\frac{3}{13} - \frac{2}{13}i$ B) $\frac{-3}{13} + \frac{2}{13}i$ C) $\frac{-3}{13} - \frac{2}{13}i$ D) $\frac{3}{13} + \frac{2}{13}i$

6. What is the length of perpendicular from the point (b, a) to the

line $\frac{x}{a} + \frac{y}{b} = 1$?

A) $\frac{b^2 - a^2 - ab}{\sqrt{a^2 + b^2}}$ B) $\frac{b^2 + a^2 + ab}{\sqrt{a^2 + b^2}}$ C) $\frac{b^2 + a^2 - ab}{\sqrt{a^2 + b^2}}$ D) $\frac{b^2 - ab + b^2}{\sqrt{b^2 - a^2}}$

7. What is the angle between the line pair represented by $2x^2 + xy - y^2 = 0$?

A) $\theta = \tan^{-1}|\sqrt{3}|$ B) $\tan^{-1}|\frac{1}{3}|$
C) $\tan^{-1}|3|$ D) $\tan^{-1}|\frac{1}{\sqrt{3}}|$

8. The roots of the equation $x^2 + 2\sqrt{3}x + 3 = 0$ are

A) real and unequal B) rational and equal
C) irrational and unequal D) irrational and equal

9. What is the value of $\lim_{x \rightarrow 2^+} \frac{\sin(x-2)}{|2-x|}$?

A) 1 B) -1 C) 0 D) does not exist

10. What is the value of k, if the function $f(x) = \begin{cases} x & \text{for } x \neq 0 \\ \frac{x}{k} & \text{for } x = 0 \end{cases}$ is continuous at $x = 0$?

A) $\frac{1}{\sqrt{3}}$ B) $\sqrt{3}$ C) 3 D) $\frac{1}{3}$

11. What is the derivative of $2(\sqrt{x} - \frac{1}{\sqrt{x}})$?

A) $\frac{x-1}{2x\sqrt{x}}$ B) $\frac{x-1}{x\sqrt{x}}$ C) $\frac{x+1}{2x\sqrt{x}}$ D) $\frac{x+1}{x\sqrt{x}}$

XI 'Science'

Subject: Com. Math

Best options among the four options. Apply tick on two options will not be counted.

Answer all the questions.

GROUP 'B'

Construct a truth table for the statement $p \vee (q \wedge r)$. State and prove De-Morgan's law.

Solve the inequality $\frac{x^2 - 2x}{x - 3} \geq 0$.

Also, show the solution in the number line.

Let $f: R - \{0\} \rightarrow R - \{2\}$ defined by $f(x) = \frac{x^2 - 2x}{x - 3}$. Find the condition inverse of f .

Can a positive number between 1 and 10 be written as the sum of two positive numbers?

Justify it by an example.

If a, b, c are in A.P., a, x, b and b, x, c are in G.P., find the value of $\frac{a}{b} \cdot \frac{b}{c} \cdot \frac{c}{a}$.

Write any two differences between real and complex numbers.

If $z = x + yi = \sqrt{\frac{1+i}{1-i}}$, show that $|z| = 1$.

Under what condition a polynomial is called a pure imaginary? Justify it by an example.

Find the condition that the lines $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ and $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represent two straight lines.

If $A = (-1, 4)$ and $B = (1, 4)$, find the equation of the line segment AB.

Find $\frac{dy}{dx}$ when $\sqrt[3]{x^2} + \sqrt[3]{y} = 1$.

Evaluate: $\lim_{x \rightarrow a} \frac{\sin 2x - \sin 2a}{\sqrt{x} - \sqrt{a}}$.

Evaluate each of the following limits to confirm your answer.

i) $\lim_{x \rightarrow 0} \frac{1}{x}$ ii) $\lim_{x \rightarrow 0} \frac{1}{x^2}$

Find the value of $f(x)$ when $f(x) = \begin{cases} 6x^2 + x - 2 & \text{if } x \neq 1 \\ \frac{5}{2} & \text{if } x = 1 \end{cases}$

Is $f(x)$ continuous at $x = 1$?

Science

Subject: Com. Mathematics (0071)

F.M.: 75

P.M.: 30

Options among the four options. Apply tick on two options will be automatically denied

all the questions.

GROUP 'B' [8×5=40]

Construct a truth table for the statement $\sim(p \vee (\sim q))$.

State and prove De-Morgan's laws for two sets.

Solve the inequality $\frac{x^2-2x}{x-3} \geq 0$.

Also, show the solution in the number line.

What condition inverse of $f(x)$ exists? Prove that the function

$f: \{0\} \rightarrow \mathbb{R} - \{2\}$ defined by $f(x) = \frac{2x+1}{x}$ is bijective. Also, find f^{-1} (if exists).

What is a positive number between two negative numbers be geometric mean?

Justify it by an example.

If a, b, c are in A.P., a, x, b and b, y, c are in G.P. Show that x^2, b^2, y^2 are in A.P.

State any two differences between real numbers and complex numbers.

If $z = x + yi = \sqrt{\frac{1+i}{1-i}}$, show that $x^2 + y^2 = 1$. Also, find $\bar{z}$.

Under what condition a pair of lines represented by $ax^2 + 2hxy + by^2 = 0$ are

imaginary? Justify it by an example.

Under the condition that the general equation of second degree

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of lines.

If $A = (-1, 4]$ and $B = [1, 3]$, find $A \cap B$ and $A - B$.

Find $\frac{dy}{dx}$ when $\sqrt[3]{x^2} + \sqrt[3]{y^2} = \sqrt[3]{a^2}$

Evaluate: $\lim_{x \rightarrow a} \frac{\sin 2x - \sin 2a}{\sqrt{x} - \sqrt{a}}$

Evaluate each of the following limits, if possible. Use a table of functional limits to confirm your conclusion.

i) $\lim_{x \rightarrow 0} \frac{1}{x}$

ii) $\lim_{x \rightarrow 0^+} \frac{1}{x}$

iii) $\lim_{x \rightarrow 0} \frac{1}{x}$

GROUP 'C' [3×8=24]

If $f(x) = \begin{cases} \frac{6x^2+x-2}{2x-1} & \text{if } x = \frac{1}{2} \\ \frac{5}{2} & \text{if } x \neq \frac{1}{2} \end{cases}$, decide whether f is continuous at $x = \frac{1}{2}$. If

f is not continuous at $x = \frac{1}{2}$, what type of discontinuity is it?

b) Define derivative of the function $f(x)$ with respect to x . By using definition, find the derivative of $\sqrt{\tan 5x}$

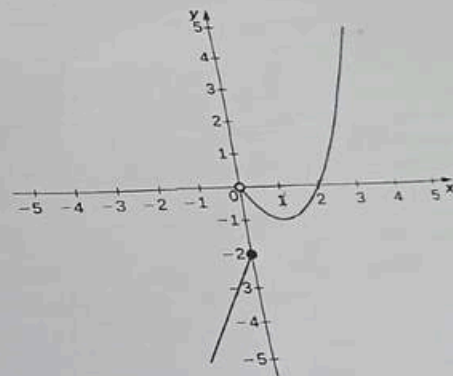
21. a) Find the domain and range of the function $f(x) = \sqrt{x^2 + x - 6}$

b) Prove that the straight lines joining the origin to the points of intersection of the line $\frac{x}{a} + \frac{y}{b} = 1$ and the curve $x^2 + y^2 = c^2$ are at right angles

if $\frac{1}{a^2} + \frac{1}{b^2} = \frac{2}{c^2}$.

22. a) If the equation $x^2 - px + q = 0$ and $x^2 - ax + b = 0$ have a common root and the other root of the second equation is the reciprocal of the other root of the first, then prove that $(q-b)^2 = bq(p-a)$.

b) Using the given graph of the function $y = f(x)$, find the answers to the following questions.



i) $\lim_{x \rightarrow 0^-} f(x)$ and $\lim_{x \rightarrow 0^+} f(x)$

ii) $\lim_{x \rightarrow 0} f(x)$

iii) $f(0)$

iv) Is $f(x)$ continuous at $x = 0$? If not continuous, what type of discontinuity is it?

v) Range of $f(x)$

"Best of Luck"

KATHMANDU MODEL SECONDARY SCHOOL
SECOND TERM EXAMINATION- 2081
SET-A

Subject: Com. Mathematics (0071)

Science'

Options among the four options. Apply tick on two options will be automatically denied

GROUP 'A' MCQ [11×1=11]

the correct option of each question in your answer sheet.

of the following compound statement is not true?

A) $(p \wedge q) \Leftrightarrow [(\sim p) \vee (\sim q)]$

B) $\sim(p \vee q) \Leftrightarrow [(\sim p) \wedge (\sim q)]$

C) $(p \Rightarrow q) \Leftrightarrow [p \vee \sim q]$

D) $p \wedge (q \vee r) \Leftrightarrow [(p \wedge q) \vee (p \wedge r)]$

is the solution of $|1 - x| > 5$?

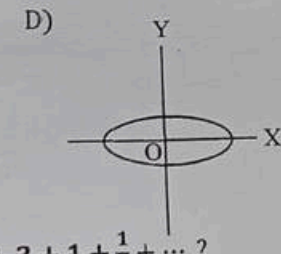
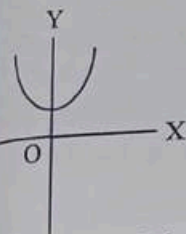
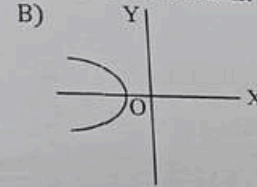
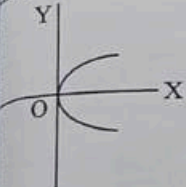
A) $(-\infty, \infty)$

B) $(4, 6)$

C) $(-\infty, -4) \cup (6, \infty)$

D) $(-\infty, 4] \cup [6, \infty)$

one of the following graph represents a function?



is the sum of the series $9 + 3 + 1 + \frac{1}{3} + \dots$?

A) $13\frac{1}{2}$

B) $\frac{25}{2}$

C) $\frac{29}{2}$

is the multiplicative inverse of $z = (-3, 2)$?

A) $-\frac{2}{13}i$

B) $\frac{-3}{13} + \frac{2}{13}i$

C) $\frac{-3}{13} - \frac{2}{13}i$

D) $\frac{-3}{13} - \frac{1}{13}i$

is the length of perpendicular from the point (b, a) to the

A) $-\frac{y}{b} = 1$?

B) $\frac{b^2 - a^2 + ab}{\sqrt{a^2 - b^2}}$

C) $\frac{b^2 + a^2 - ab}{\sqrt{a^2 + b^2}}$

D) $\frac{b^2 - ab + b^2}{\sqrt{b^2 - a^2}}$

is the angle between the line pair represented by $2x^2 - xy - y^2 = 0$?

A) $\theta = \tan^{-1}|\sqrt{3}|$

B) $\tan^{-1}|\frac{1}{3}|$

C) $\tan^{-1}|3|$

D) $\tan^{-1}|\frac{1}{\sqrt{3}}|$

8. The roots of the equation $x^2 + 2\sqrt{3}x + 3 = 0$ are

A) real and unequal

B) rational and equal

C) irrational and equal

D) irrational and unequal

9. What is the value of $\lim_{x \rightarrow 1^-} \frac{\sin(x-1)}{|1-x|}$?

A) 1

B) -1

C) 0

D) does not exist

10. What is the value of k, if the function $f(x) = \begin{cases} \frac{x}{e^{2x}-1} & \text{for } x \neq 0 \\ k & \text{for } x = 0 \end{cases}$ is continuous at $x = 0$?

A) $\frac{1}{\sqrt{2}}$

B) $\sqrt{2}$

C) 2

D) $\frac{1}{2}$

11. What is the derivative of $2(\sqrt{x} + \frac{1}{\sqrt{x}})$?

A) $\frac{x-1}{2x\sqrt{x}}$

B) $\frac{x-1}{x\sqrt{x}}$

C) $\frac{x+1}{2x\sqrt{x}}$

D) $\frac{x+1}{x\sqrt{x}}$

वशानुगत रूपमा पनि मानसिक रोग लाग्ने सम्भावना देखिन्छ। मानसिक रोगको एउटा प्रमुख कारण पारिवारिक भगडा, बेरोजगारी, प्राकृतिक प्रकोप भएको हुनाले विरामी हुन नदिन र विरामी भएकालाई छिटो निको पार्ने परिवारका सदस्यको भूमिका ठूलो हुन्छ। परिवार सामु देखिने जटिल समस्या एक आपसमा मिलेर समाधान गर्ने सकिने हुनाले उक्त भोग एउटामा मात्र पर्न दिनु हुँदैन। विरामीलाई डाक्टरको सल्लाह बमोजिम नियमित औषधि सेवन गराउनुपर्छ। विरामी रिसाउने, फर्किने तथा अन्य भगडालु स्वभाव देखाउन थालेमा आफू उत्तेजित हुनुहुँदैन। विरामीले रोगको कारणले त्यस्तो व्यवहार देखाएको हो भन्ने बुझ्नुपर्छ। औषधिले काम गर्न कमिमा एक हप्ता लाग्छ, त्यसैले तत्काल औषधिको असर खोज्नु हुँदैन। नेपालमा रक्सी र लागूऔषधको समस्या बढ्दै गएको छ। यसका लागि पारिवारिक सदस्यको रेखदेख र नियन्त्रण अति आवश्यक छ। साथै समुदायमा बस्ने मानिस एकजुट भई काम गरे यो नियन्त्रण गर्न सकिन्छ।

प्रश्नहरू :

क) विराममा कुनै न कुनै प्रकारको मनोरोगबाट पीडित भएका मानिसको सङ्ख्या कति छ ?
 अ) करीब २० करोड आ) करीब २८ करोड इ) करीब ४ करोड
 ब) विराममा कतिवटा खतरनाक रोगले मानिसको दैनिक क्रियाकलापमा असर गर्छ ?
 अ) ४ वटा आ) १५ वटा इ) १० वटा ई) २० वटा

ग) मानसिक रोग लाग्नुका प्रमुख कारण के हुन् ?

अ) पारिवारिक असन्तुष्टि, बेरोजगारी, प्राकृतिक असन्तुलन
 आ) पारिवारिक चिन्ता, अल्छीपना
 इ) पारिवारिक बेमेल, बेरोजगारी, जलवायु परिवर्तन
 ई) पारिवारिक भगडा, बेरोजगारी, प्राकृतिक प्रकोप

घ) मानसिक विरामी भएकालाई छिटो निको पार्न कसको भूमिका ठूलो हुन्छ ?

अ) परिवारको आ) श्रीमान र श्रीमतीको
 इ) छोरा र छोरीको ई) आफ्नै

१०. दिइएको वनस्पतिबाट चारकोटा बुँदा टिपी एक तृतीयांशमा सारांश लेख्नुहोस्: २+२=४

नेपालको भूमिमा प्रशस्त खनिज सम्पदाहरू छन्। नेपाली भूगोलको सतह तथा भूगर्भभित्र सुन, चाँदी, फलाम, तामा, अभ्रक, सिसा, जस्ता, खरी, स्लेट, मार्बल, गेरु, कोइला, ग्याँस, पेट्रोलियम, युरेनियम, सोडा, बिरेनियम आदि अनेकौं खनिजहरू छन्। सुनकोसी, बुढी गण्डकी, काली गण्डकी, रिउ खोला, कोसी आदिमा सुन पाइन्छ। यसका साथै कर्नालीको बगरमा पनि सुन पाइन्छ। चिसापानी गढी, बागलुङ र फूलचोकमा चाँदी पाइन्छ भने ठोसे, फूलचोक, चितवन, कुलेखानी, भैँसे, फर्पिङ, डोटी आदिमा प्रशस्त फलाम पाइन्छ। तामाखानी, कुलेखानी, बुढ खोला, वाप्सा, मार्फा, इलाम, म्याग्दी आदिमा तामा र भोजपुर, सिन्धुली, लमजुङ, गोसाइँकुण्ड, सुन्दरीजल जस्ता ठाउँमा अभ्रक भन्ने खनिज पाइन्छ। चोभार, भैँसे, धनकुटा, मोरङ जस्ता ठाउँमा चुनडुङ्गा र गणेश हिमाल, फूलचोक, मकवानपुर, माजेर खोला आदि ठाउँमा सिसा जस्ता पाइन्छन्। त्यसै गरी चिसापानी गढी, गोसाइँकुण्डमा गन्धक तथा खोटाङ, सिन्धुपाल्चोक, भोजपुरमा जस्ता पाइन्छन्। दाङ, सल्यान, चौतारा, काठमाडौँ उपत्यका आदि ठाउँमा कोइला र नेपालगन्ज, मुक्तिनाथ, दैलेख, प्युठान, कोइलाबास, दाङलगायतका धेरै ठाउँमा पेट्रोलियम पाइने सम्भावना छ। अत्यन्त बहुमूल्य मानिने युरेनियम मुक्तिनाथलगायतका केही ठाउँमा पाइने कुरा अनुसन्धानहरूले देखाएको छ। यस्ता खनिजहरूको आधुनिक तरिकाले उत्खनन, प्रशोधन र प्रयोग गर्न आवश्यक छ।

११. कुनै एक प्रश्नको उत्तर दिनुहोस्:

क) धमला वस्त्रालय, वानेश्वरले गिल्सयाण्ड वस्त्रालय, न्युरोडलाई दशैँका लागि आवश्यक पर्ने विभिन्न डिजाइनका कपडाहरू पठाईदिन आग्रह गर्दै लेखिने व्यावसायिक पत्रको तमुना लेख्नुहोस्।

ख) विद्युतीय कारको बजार प्रवर्धनका लागि बनाइने विज्ञापनको सचित्र आकर्षक नमुना बनाउनुहोस्।

१२. कुनै एक प्रश्नको उत्तर दिनुहोस्:

क) केएमसी विद्यालयले आफ्नो पञ्जीसी वार्षिक उत्सव कार्यक्रममा आयोजना गरेको भित्री म्याराथन प्रतियोगिता का सम्पूर्ण कार्यक्रमलाई समेटेर १५० शब्दसम्मको प्रतिवेदन लेख्नुहोस्।
 ख) 'मूल्यवृद्धि'ले निम्त्याएका समस्याहरू विषयमा १५० शब्दसम्मको टिप्पणी लेख्नुहोस्।

१३. छउपडी प्रथा बन्त्य गर्न के गर्नुपर्ला ? १२५ शब्दसम्मको जवाफ लेख्नुहोस्।

१४. तलका कुनै दुई प्रश्नको उत्तर दिनुहोस्:

क) दिइएको कवितामा पढी सोधिएका प्रश्नको उत्तर दिनुहोस्:
 कुन पहाडले कुन खोलाले तिमी गाँत छेकेछ्यो र
 वीर पुर्खा ! कुन ओढीले तिमी यात्रा रोकेछ्यो र
 गरुडको भैंँ वेग तिमी कुन आकाशले बाँध्न सज्यो
 पौरखले रच्यो नेपाल ! पहाड तराई जुट्न सज्यो।

प्रश्नहरू:

क) नेपालीहरूको कस्तो पौरखले नेपालको रचना भएको हो ?
 ख) कवितामा आधारमा हामी वीर पुर्खाको गौरव गाथाको वर्णन गर्नुहोस्।

ख) दिइएको कविता पढी सोधिएका प्रश्नहरूको उत्तर लेख्नुहोस्:

खोला अचल फेरि गाउँतिर सोधिएको रहेछ। यो खोलाले पनि गाउँलाई धुक्के रुवायो। हुन न रुवायो भन्नु खोलाप्रति अन्याय ठहर्ला। यो खोलाले गाउँलाई पालेको पनि छ। हाम्रो विषाणको लक्ष्यमा भन्नु उमेर यही खोलाको हिल र बगरहरूमा बितेको थियो। कहिले ओरहा खाने निहुँमा त कहिले गाईबैसी चराउने निहुँमा। खोलाको किनारमा हामी गाईको पुच्छर समाएर बाढी तने गर्छौं अथवा यस्तो बेलामा हामी खोलाको किनारमा बसेर बाढीको आनन्द लुट्छौं। तर्न छराउने कोही पाँचै आए भने हामी तीनचार जना एकअर्काको हात समाउँदै बाढीमा हेलिन्छौं र किनारमा पाँचैको मानिसलाई तारिदिन्छौं र यस्तो काममा खास गरी परिछन्ने विशेष रुचि थियो। "यहीनिर कल छ, यो सालको वर्षामा मूल फुटेको एकदम चिसो र मिठो पानी छ।" ऊ पानी लिएर आइसकेको थियो।

प्रश्नहरू:

अ) खोलाले रुवायो भन्ने कुरा मात्र गर्दा कसरी अन्याय हुन्छ ?
 आ) 'म' पात्रका खोलासँगका बालअनुभूति कस्ता छन् ?

ग) दिइएको चिठीको अंश पढी प्रश्नहरूको उत्तर दिनुहोस्:

मित्र सुरज ! अवश्य नै पढाइ छोडेर धन कमाउन परदेशिने मेरो निर्णय अत्यन्त गलत थियो। धन कमाउन नसके पनि आफ्नो गाउँ, आफ्नो परिवार, इष्टमित्र, समाज र देश सम्भरेर घर फर्कने मेरो निर्णय गलत हुने छैन। त्यसो त निर्णय आफैँमा सही वा गलत हुँदैन। त्यसलाई सही वा गलत बनाउने त हामी आफैँले हो। त्यसैले घर फर्कने मेरो यो निर्णयलाई म जसरी पनि सही साबित गर्छु।

प्रश्नहरू :

क) म पात्रले गरेका सही र गलत निर्णय के के थिए ?
 ख) म पात्रको घर फर्कने निर्णयप्रति तपाईंको धारणा लेख्नुहोस्।

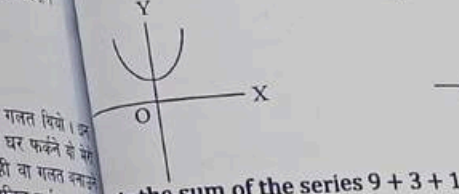
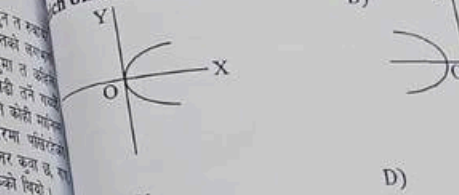
१५. तलका मध्ये कुनै एक प्रश्नको उत्तर दिनुहोस्।

क) योगमायाले धर्मराज्यको स्थापनाको लागि राखेका मागहरू केके हुन् र ती मागहरू अहिलेको समयमा कतिको सान्दर्भिक छन् ? योगमायाको जीवनीका आधारमा समीक्षा गर्नुहोस्।
 ख) 'त्यो फेरि फर्कला ?' कथाका आधारमा समीक्षा गर्नुहोस्।
 छ) 'त्यो फेरि फर्कला ?' कथाका आधारमा समीक्षा गर्नुहोस्।
 १६. तल दिइएका मध्ये कुनै एक शीर्षकमा २५० शब्दसम्ममा नघटाई निबन्ध लेख्नुहोस्।
 क) मभिन्नको देशप्रेम
 ग) सामाजिक जागरणमा शिक्षा र सञ्चार

Subject: Com. Mathematics (0071)
 Time: 5 min.
 Give options among the four options. Apply tick on two options will be counted.

GROUP 'A' MCQ [11×1=11]

Choose the correct option of each question in your answer sheet. Each of the following compound statement is not true. Which one of the following is the solution of $|1-x| > 5$?



is the sum of the series $9 + 3 + 1 + \dots$
 B) $13\frac{1}{2}$
 is the multiplicative inverse of z
 C) $-\frac{2}{13}$
 is the length of perpendicular from the origin to the line $\frac{x}{a} + \frac{y}{b} = 1$
 D) $\frac{b^2 - a^2 + ab}{\sqrt{a^2 + b^2}}$
 is the angle between the line